

HDOC-10

Question 1:

Write a C program to input time in seconds and convert it into days, hours, minutes, and seconds format.

Task1: algorithm:

Step1: Start.

Step2: Declare variables totalSeconds, remainingSeconds, days, hours, minutes, and seconds.

Step3: Input the total time in seconds.

Step4: Calculate days using:

$$\text{days} = \text{totalSeconds} / 86400$$

Step5: Find the remaining seconds:

$$\text{remainingSeconds} = \text{totalSeconds} \% 86400$$

Step6: Calculate hours:

$$\text{hours} = \text{remainingSeconds} / 3600$$

Step7: Update remaining seconds:

$$\text{remainingSeconds} = \text{remainingSeconds} \% 3600$$

Step8: Calculate minutes:

$$\text{minutes} = \text{remainingSeconds} / 60$$

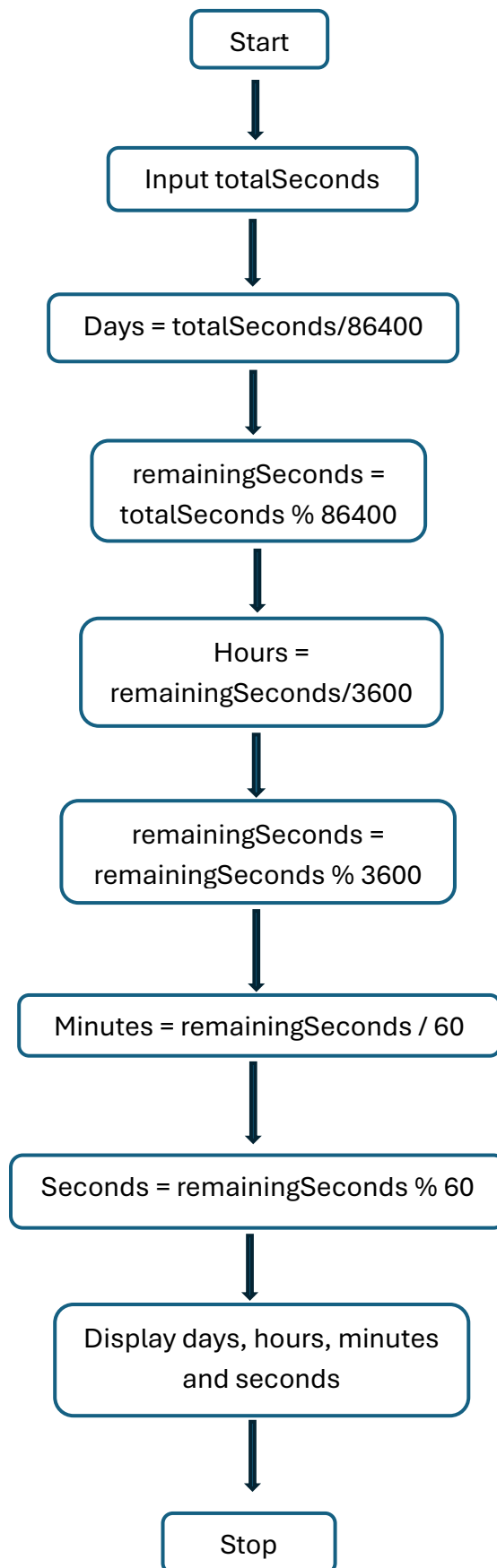
Step9: Calculate seconds:

$$\text{seconds} = \text{remainingSeconds} \% 60$$

Step10: Display days, hours, minutes, and seconds.

Step11: Stop.

Task2: flowchart:



Task3: Pseudocode:

- START
- INPUT totalSeconds
- $\text{days} = \text{totalSeconds} / 86400$
 $\text{remainingSeconds} = \text{totalSeconds} \% 86400$
- $\text{hours} = \text{remainingSeconds} / 3600$
 $\text{remainingSeconds} = \text{remainingSeconds} \% 3600$
- $\text{minutes} = \text{remainingSeconds} / 60$
 $\text{seconds} = \text{remainingSeconds} \% 60$
- DISPLAY days
- DISPLAY hours
- DISPLAY minutes
- DISPLAY seconds
- STOP

Task4: Test case table:

Test Case	Input (seconds)	Expected Output	Actual Output	Result
1	86400	1 day, 0 hour, 0 minute, 0 second	1 day, 0 hour, 0 minute, 0 second	PASS
2	90061	1 day, 1 hours, 1 minutes, 1 seconds	1 day, 1 hours, 1 minutes, 1 seconds	PASS
3	2355786	27 days, 6 hour, 23 minute, 6 second	27 days, 6 hour, 23 minute, 6 second	PASS
4	180	0 days, 0 hours, 3 minute, 0 seconds	0 days, 0 hours, 3 minute, 0 seconds	PASS

Task5: C program:

```
#include <stdio.h>
int main()
{
    int totalSeconds;
    int remainingSeconds;
    int days, hours, minutes, seconds;

    printf("Enter time in seconds: ");
    scanf("%d", &totalSeconds);
```

```

days = totalSeconds / 86400;
remainingSeconds = totalSeconds % 86400;

hours = remainingSeconds / 3600;
remainingSeconds = remainingSeconds % 3600;

minutes = remainingSeconds / 60;
seconds = remainingSeconds % 60;

printf("\nDays = %d", days);
printf("\nHours = %d", hours);
printf("\nMinutes = %d", minutes);
printf("\nSeconds = %d\n", seconds);
return 0;
}

```

Task6: output:

```

(kali㉿kali)-[~/HDOC9.c/HDOC10.c]
$ ./a.out
Enter time in seconds: 86400

Days = 1
Hours = 0
minutes = 0
seconds = 0

```

```

(kali㉿kali)-[~/HDOC9.c/HDOC10.c]
$ ./a.out
Enter time in seconds: 90061

Days = 1
Hours = 1
minutes = 1
seconds = 1

```

```

(kali㉿kali)-[~/HDOC9.c/HDOC10.c]
$ ./a.out
Enter time in seconds: 2355786

Days = 27
Hours = 6
minutes = 23
seconds = 6

```

```

(kali㉿kali)-[~/HDOC9.c/HDOC10.c]
$ ./a.out
Enter time in seconds: 180

Days = 0
Hours = 0
minutes = 3
seconds = 0

```

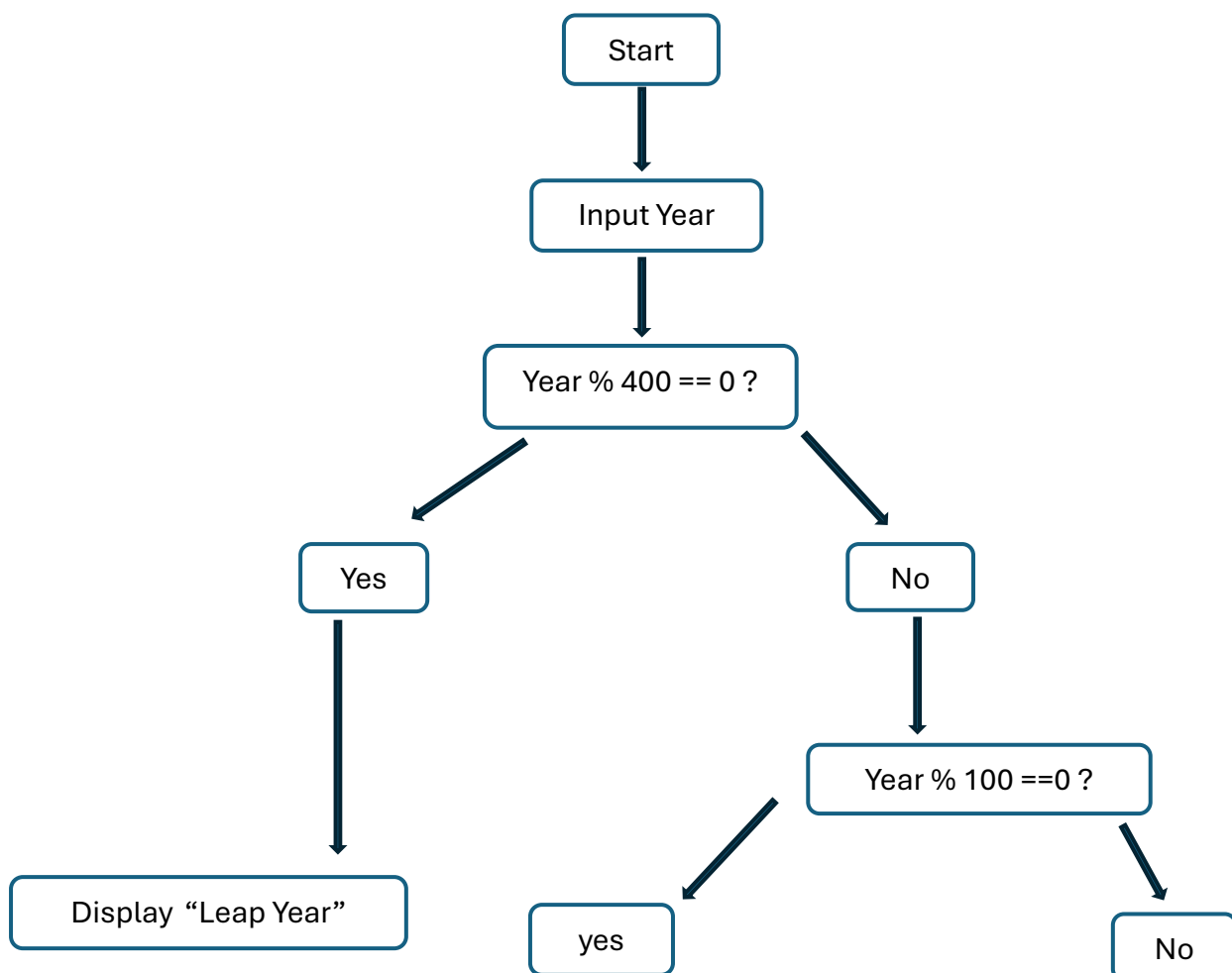
Question 2:

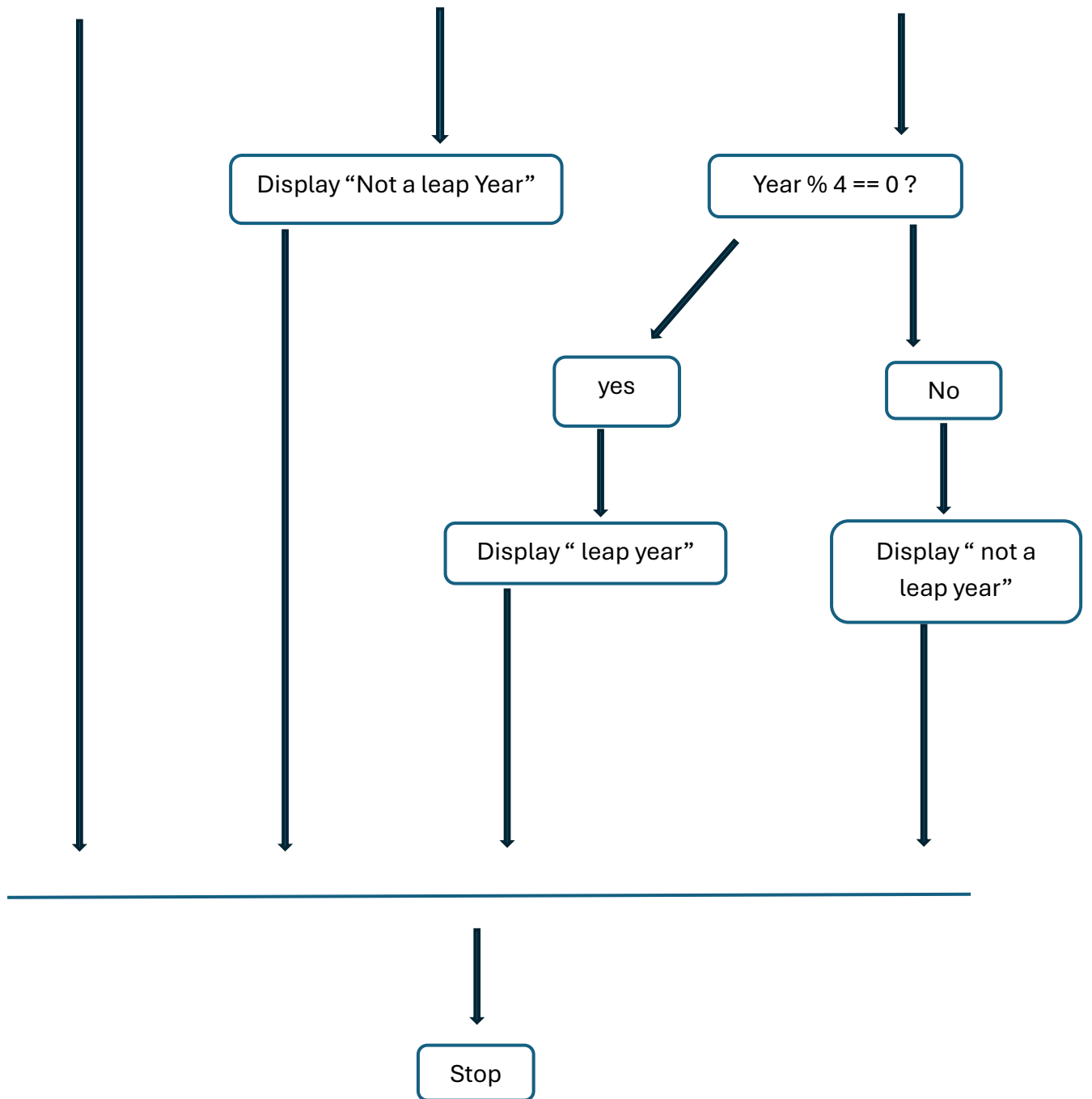
Write a program to input a year and check whether it is a leap year or not.

Task1: algorithm:

- Start
- Input the year as year.
- Check if the year is divisible by 400.
If yes, it is a leap year.
- Otherwise, check if the year is divisible by 100.
If yes, it is not a leap year.
- Otherwise, check if the year is divisible by 4.
If yes, it is a leap year.
- Otherwise, it is not a leap year.
- Display the result.
- Stop.

Task2: flowchart:





Task3: Pseudocode:

```
START

INPUT year

IF year % 400 == 0 THEN
    DISPLAY "Leap Year"

ELSE IF year % 100 == 0 THEN
    DISPLAY "Not a Leap Year"

ELSE IF year % 4 == 0 THEN
    DISPLAY "Leap Year"

ELSE
    DISPLAY "Not a Leap Year"

END IF

STOP
```

Task4: Test case table:

Test Case	Input Year	Expected Output	Actual Output	Result
1	2008	Leap Year	Leap Year	PASS
2	1948	Leap Year	Leap Year	PASS
3	2019	Not a Leap Year	Not a Leap Year	PASS
4	2026	Not a Leap Year	Not a Leap Year	PASS

Task5: C program:

```
#include <stdio.h>
int main()
{
    int year;
    printf("Enter a year: ");
    scanf("%d", &year);

    if (year % 400 == 0)
    {
        printf("%d is a leap year.\n", year);
    }
    else if (year % 100 == 0)
    {
        printf("%d is not a leap year.\n", year);
    }
    else if (year % 4 == 0)
    {
        printf("%d is a leap year.\n", year);
    }
    else
    {
        printf("%d is not a leap year.\n", year);
    }
    return 0;
}
```

Task6: Output:



```
(kali㉿kali)-[~/HDOC9.c/HDOC10.c]
$ cc ques2.c

(kali㉿kali)-[~/HDOC9.c/HDOC10.c]
$ ./a.out
Enter a year: 2008
2008 is a leap year.
```

```
(kali㉿kali)-[~/HDOC9.c/HDOC10.c]  
$ ./a.out  
Enter a year: 1948  
1948 is a leap year.
```

```
(kali㉿kali)-[~/HDOC9.c/HDOC10.c]  
$ ./a.out  
Enter a year: 2019  
2019 is not a leap year.
```

```
(kali㉿kali)-[~/HDOC9.c/HDOC10.c]  
$ ./a.out  
Enter a year: 2026  
2026 is not a leap year.
```